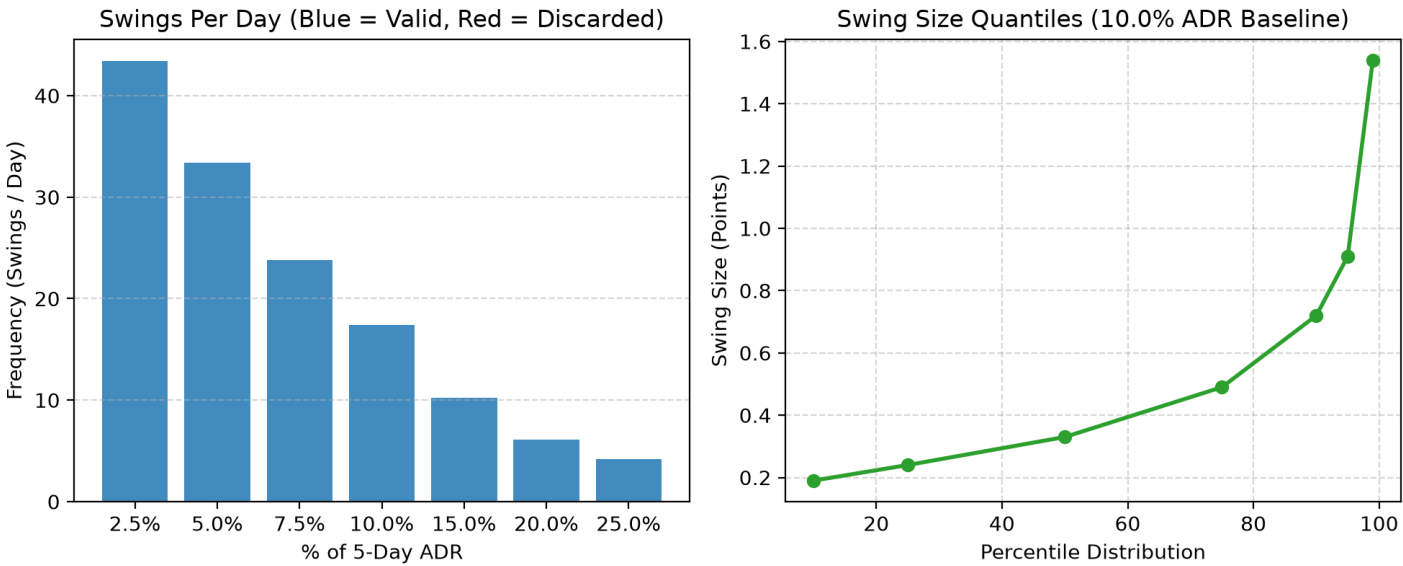


Statistical Swing Analysis Report: VALE3 (5min)

5-Day Average Daily Range (ADR_5d): 1.66 pts

VALE3 (5min) - Swing Distribution & Sensitivity Analysis



Parameter Sweep & Noise Filtering Matrix

Pct ADR	Threshold Pts	Total Swings	Swings Per Day	Avg Bars Per Swing	Mean Size	Median Size	P75 Size	P90 Size	P95 Size	Status
2.5%	0.04	21634	43.4	3.0	0.22	0.16	0.26	0.44	0.6	VALID
5.0%	0.08	16644	33.4	3.6	0.27	0.2	0.32	0.5	0.68	VALID
7.5%	0.12	11837	23.8	4.7	0.34	0.27	0.4	0.6	0.79	VALID
10.0%	0.17	8673	17.4	6.0	0.41	0.33	0.49	0.72	0.91	VALID
15.0%	0.25	5079	10.2	9.5	0.56	0.45	0.66	0.96	1.23	VALID
20.0%	0.33	3058	6.1	15.2	0.75	0.6	0.88	1.28	1.59	VALID
25.0%	0.41	2084	4.2	21.8	0.92	0.75	1.1	1.56	1.98	VALID